

Vivaan Singhvi

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EDUCATION

University of Michigan | Ann Arbor, MI

August 2024 – Expected 2026

- GPA: 4.0
- Major: Computer Science, College of Engineering
- Relevant Coursework: Foundations of LLMs, Introduction to Machine Learning, GPU Programming
- Peer Advisor for Michigan Research and Discovery Scholars (MRADS)

SELECT EXPERIENCE

Robust Adversarial Patch Generation | Oak Ridge National Laboratory

May 2025 – July 2025

- Created an end-to-end system to make robust patch attacks against YOLO models from input images
- Explored dozens of 3D reconstruction algorithms, including variations on NeRFs and Gaussian Splatting
- Built an automated mesh reconstruction pipeline, handling meshes with over 5 million vertices
- Achieved over 95% attack success on target objects digitally and around 75% in the real world

Phylogenetic Reconstruction Optimization | University of Michigan

September 2024 – Present

- Developed an optimized C++ phylogenetic reconstruction algorithm with over 100-fold speedup
- Enabled super-scale phyloanalysis, demonstrated by building a billion-tip tree in under 3 hours
- Gained experience with optimization techniques in algorithm design and implementation
- Published and presented a paper on the work at the ALIFE 2025 conference in Kyoto, Japan

Adversarial Image Detection | Oak Ridge National Laboratory

June 2024 – July 2024

- Formed a deep understanding of neural networks and the AI framework PyTorch
- Built a solid foundation in adversarial artificial intelligence, learning about several attacks and defenses
- Created a robust system to detect basic adversarial images with over 99% accuracy
- Presented work in front of about 30 people in a 5-minute lightning talk format
- Developed a novel attack completely invisible to the human eye, resistant to image transformations

SELECT PROJECTS

DreamBerd Interpreter | Python

- Developed the first near-complete interpreter for the fictional language DreamBerd
- Won a \$99 bounty for a charity of my choice and earned nearly 400 GitHub stars
- Gained in-depth knowledge about programming language design and implementation
- Used GitHub actions to implement a server where people could share global variables across runs

Rubik's Cube Solver | Python, JavaScript, WebSockets

- Created a Python package to model and solve a Rubik's cube of any size
- Applied OpenCV and mathematical algorithms to read a cube from a single video
- Bootstrapped a web application, featuring an interactive and editable cube demonstrating the process
- Integrated WebSockets for a fast user experience, communicating video, cube state, and solution data

SKILLS

Languages: English (Fluent), Hindi (Verbal), German (Beginner)

Programming Languages: Python, C, C++, JavaScript/TypeScript, HTML/CSS, SQL, Java

Libraries: PyTorch, OpenCV, PaddleOCR, Scikit-Learn, React, NumPy, Pandas, Tensorflow

Tools: Git/GitHub, MacOS, Unix Shell, WSL, Neovim, VS Code, Vim, PostgreSQL